

Multi-Region Variance-Hierarchy Guided Neural Steering

Deriving steering vectors for non-dominant latent senses — unsupervised.

Scope. A self-contained technical brief: the problem, why existing methods fall short, and the methodology with supporting math. Final novelty and obviousness calls are for patent counsel; the positioning section is engineering rationale, not a legal opinion.

1 Problem statement

A language model represents each token or context as a vector — an **activation** $h \in \mathbb{R}^d$. *Steering* means adding a chosen direction to that vector at inference time,

$$h' = h + \alpha \cdot \hat{d}, \quad \hat{d} = d / \|d\|, \quad \alpha \in \mathbb{R}$$

so the model's output leans toward ($\alpha > 0$) or away from ($\alpha < 0$) a target concept, sense, or behavior. The object we need is the **steering vector** d .

Derive a steering vector for a concept that is **not** the dominant pattern in the activations — a sense or behavior that is rare, masked by a louder one, split across several regions, or diffuse — and do so **without a large labeled contrastive dataset**.

Why this matters. The interesting and safety-relevant senses are frequently the *minority* ones. A word's common meaning dominates its activations while a rarer meaning hides underneath: *bank* (institution $\gg$ river), *shoot* (photograph $\gg$ firearm), *crack* (fissure $\gg$ narcotic), *fire* (warmth $\gg$ weapon). The dangerous reading is usually the quiet one, so any method that recovers only the loud reading is structurally blind to exactly the cases that matter most.

Formal goal. Given activations $H \in \mathbb{R}^{n \times d}$ collected for a target token or concept, identify direction(s) d that isolate a target latent sub-population — a "region" — **even when that region carries little variance**, using at most a handful of orienting exemplars rather than a supervised training set.

PROBLEM

Steer toward a meaning that is not the dominant one

A common sense drowns out a rare one; we still need a clean push toward the rare sense.

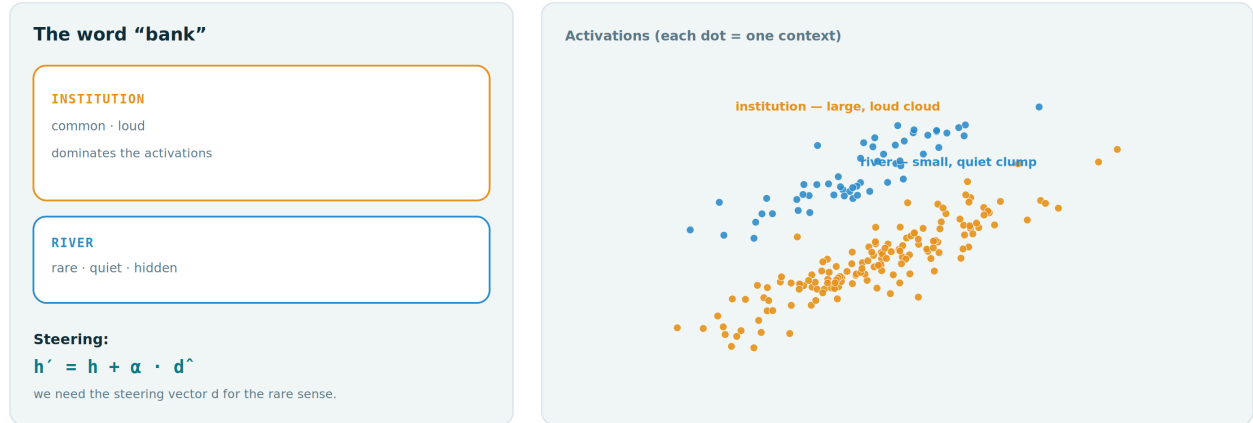


Figure 1. The setup: a common sense dominates the activation cloud while a rare sense hides as a small clump; we still need a clean steering push toward the rare sense.

2 Notation

Symbol	Meaning
$\mathbf{H} \in \mathbb{R}^{n \times d}$	activation matrix; n contexts, hidden dim d
$\mu, \mathbf{H}_c$	mean activation; centered $\mathbf{H}$ (rows $\mathbf{h}_i - \mu$)
$\Sigma = (1/n) \mathbf{H}_c^T \mathbf{H}_c$	covariance; eigendecomposition $\Sigma = \sum_k \lambda_k \mathbf{v}_k \mathbf{v}_k^T, \lambda_0 \geq \lambda_1 \geq \dots$
$\mathbf{V}_r = [\mathbf{v}_0 \dots \mathbf{v}_{r-1}]$	top- r ("dominant") eigenvector subspace
$\Delta = \mu_B - \mu_A$	difference of sub-cluster means
$z_i(\mathbf{d}) = \mathbf{d}^T \tilde{\mathbf{h}}_i$	1-D projection of point i onto direction $\mathbf{d}$

3 Shortcomings of current methods

All mainstream approaches are biased toward the **dominant** direction, or require supervision the hidden sense does not have.

3.1 Difference-of-means / mean-activation steering (CAA, ActAdd-style). Compute $\mathbf{d} = \mu_+ - \mu_-$ over contrast sets. Because the steering vector is an *average*, a minority sub-population is averaged away: if 90% of "bank" contexts are the institution sense, μ is essentially the institution centroid and the river sense never moves it. Also requires curated contrast pairs.

3.2 Top principal component. PCA returns the directions of **maximum variance**, on the false assumption that *variance* = *semantic salience*. A minority sense is, by construction, low-

variance — yet can be perfectly well-separated. PCA spends its top component on the loud axis and never surfaces the quiet one.

The crux: **the hidden sense is not a high-variance direction. It is a low-variance direction whose projection is bimodal.** Variance and multimodality are different objectives, and every variance-driven method optimizes the wrong one.

3.3 Supervised probes / LDA. A discriminant $d \propto S_W^{-1} \Delta$ cleanly separates two classes — *if you have labels for them*. By assumption the hidden sense is unlabeled (that is what makes it hidden), and LDA also requires inverting a $d \times d$ within-class scatter, unstable at LLM hidden dimensions.

3.4 Dictionary learning / SAEs. Powerful but heavyweight: they require training a large auxiliary model, the learned features are not guaranteed to align with the *specific* concept to steer, and extracting a single targeted minority sense as a usable vector is indirect.

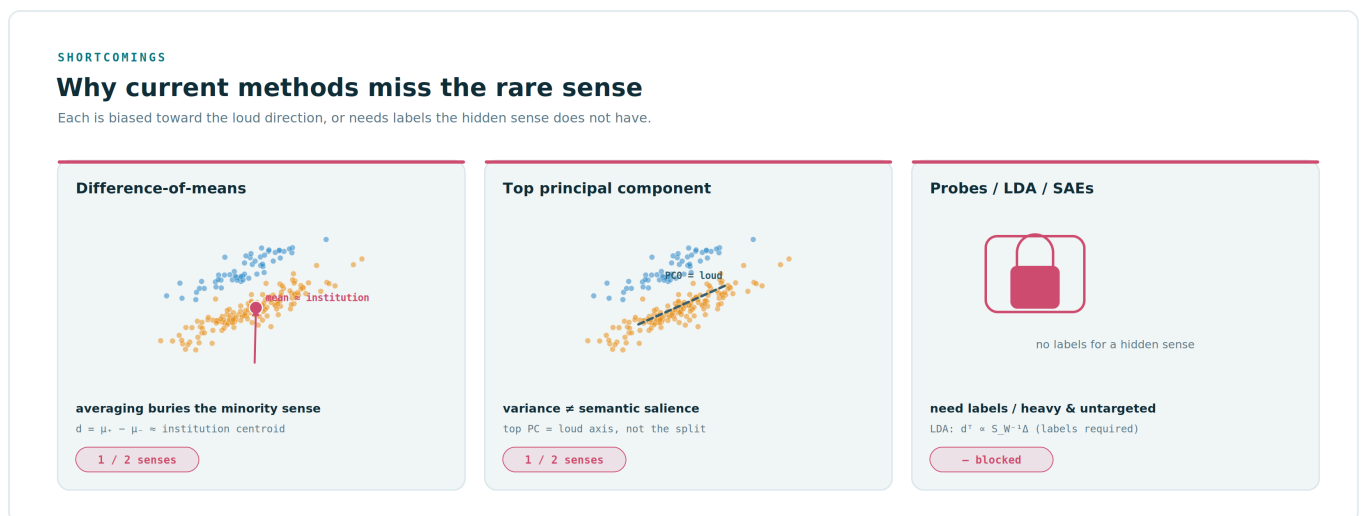


Figure 2. Three failure modes — averaging buries the minority sense; variance $\neq$ salience; discriminant and dictionary methods need labels the hidden sense lacks.

CORE INSIGHT

Variance and bimodality are different objectives

The hidden sense is a low-variance direction whose projection splits into two.

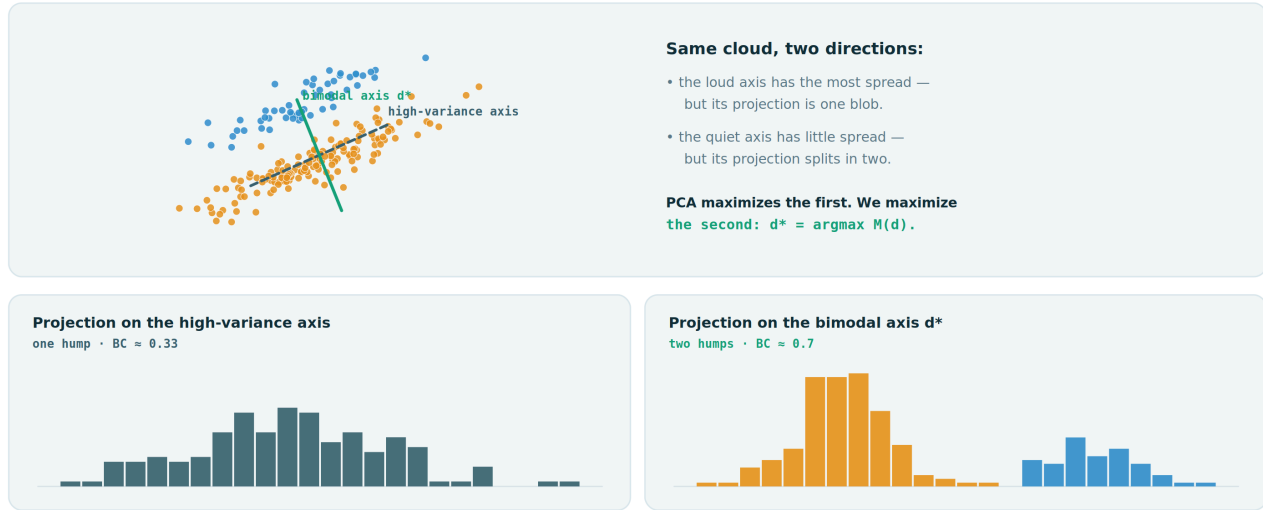


Figure 3. The core insight. Same cloud, two directions: the high-variance axis projects to one hump (BC ≈ 0.33); the bimodal axis d^* projects to two (BC ≈ 0.7).

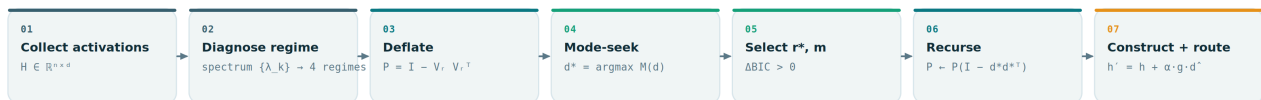
4 Methodology

Two moves, applied recursively: **deflate** the dominant directions, then **mode-seek** for the bimodal split in what remains. A diagnosis stage routes each target into one of four regimes, and a selection rule decides how many directions to remove and how many senses to extract.

METHOD · END-TO-END

The variance-hierarchy guided steering pipeline

Diagnose the regime, deflate the loud subspace, mode-seek the hidden split, then recurse and route.



Steps 03–06 repeat: each pass peels off one hidden sense, building an ordered steering subspace $D = [d^*, d^*, \dots]$. Step 02 routes Masked / Multi-region / Diffuse targets — the cases ordinary methods miss.

Figure 4. The end-to-end pipeline. Steps 03–06 repeat, peeling off one hidden sense per pass to build an ordered steering subspace.

4.1 Variance-hierarchy diagnosis (regime routing)

From the spectrum $\{\lambda_k\}$ and the residual structure, classify the target concept as **Dominant** (the target sense is the top component — ordinary methods suffice), **Masked** (a louder unrelated factor sits on top — deflate it, then recover), **Multi-region** (several disjoint clusters — recover a steering subspace), or **Diffuse** (spread thinly across many low-variance directions — aggregate weak evidence). The pipeline below handles Masked / Multi-region / Diffuse, where existing methods fail.

DIAGNOSIS

Four regimes the diagnosis routes between

The pipeline targets Masked, Multi-region, and Diffuse — where ordinary methods fail.

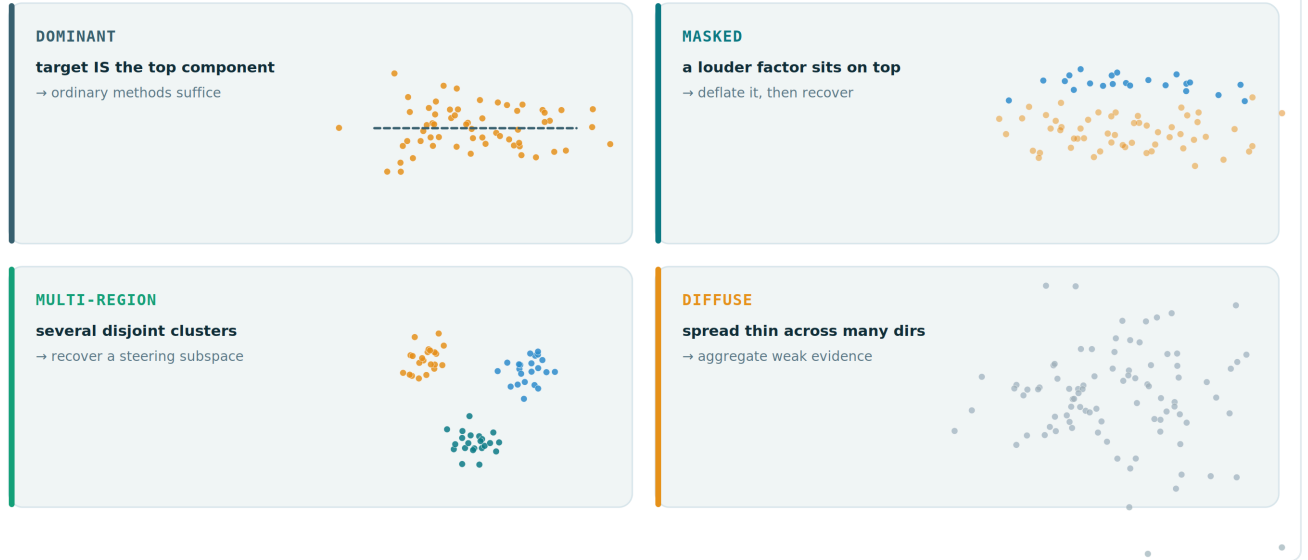


Figure 5. The four regimes the diagnosis routes between.

4.2 Deflation — suppress the dominant subspace

Form the orthogonal projector onto the complement of the top- r directions and apply it:

$$\begin{aligned} P &= I_d - V_r V_r^T && \text{(deflation projector)} \\ \tilde{H} &= H_c P && \text{(residual activations, } \tilde{h}_i = P(h_i - \mu)) \end{aligned}$$

P removes the loud variance so it can no longer dominate the search. Geometrically, the activation cloud is a long "cigar" along the loud axis; P collapses that axis and leaves the **residual subspace** — the place the hidden structure lives. Mode-seeking (4.3) runs entirely inside $\text{range}(P)$: in a 3-D intuition, one loud axis is removed and the search rotates within the remaining 2-D plane.

4.3 Mode-seeking — projection pursuit for the hidden split

We do **not** look for the direction of maximum variance. We look for the direction whose 1-D projection is **maximally bimodal**:

$$d^* = \underset{\|d\|=1, d \in \text{range}(P)}{\text{argmax}} \quad M(d)$$

where $M(d)$ is a multimodality index of the projected sample $\{z_i(d) = d^T \tilde{h}_i\}$. Three interchangeable choices, in increasing rigor:

(a) Sarle's bimodality coefficient — cheap, closed form. With sample skewness g_1 and excess kurtosis g_2 :

$$BC(d) = \frac{g_1^2 + 1}{g_2^2 + 1} \quad (\approx 0.33 \text{ for one hump ; } \rightarrow 1 \text{ for two ; bimodal when } BC \rightarrow 1)$$

(b) Hartigan's dip statistic — a nonparametric, robust test of departure from unimodality.

(c) GMM evidence gap (*preferred — reuses existing batched-EM / GMM infrastructure*). Fit a 1-component and a 2-component Gaussian mixture to $\{z_i\}$ and compare evidence by BIC:

$$\begin{aligned} \text{BIC}_k &= -2 \log \hat{L}_k + k_{\text{params}} \cdot \log n & (k_{\text{params}}: 2 \text{ for 1-comp, } 5 \text{ for 2-comp}) \\ \Delta\text{BIC}(d) &= \text{BIC}_1 - \text{BIC}_2 & (> 0 \Rightarrow \text{two modes favored}) \\ M(d) &= \Delta\text{BIC}(d) \end{aligned}$$

The evidence gap is attractive because the **same statistic** that finds the direction also decides whether a second mode exists at all, and supplies the stopping rule in 4.5 — no separate threshold heuristic. Because the residual is low intrinsic dimension, the search is cheap: a FastICA-style fixed-point on the unit sphere, or grid-then-refine. The recovered d^* is, by construction, orthogonal to the deflated loud subspace.

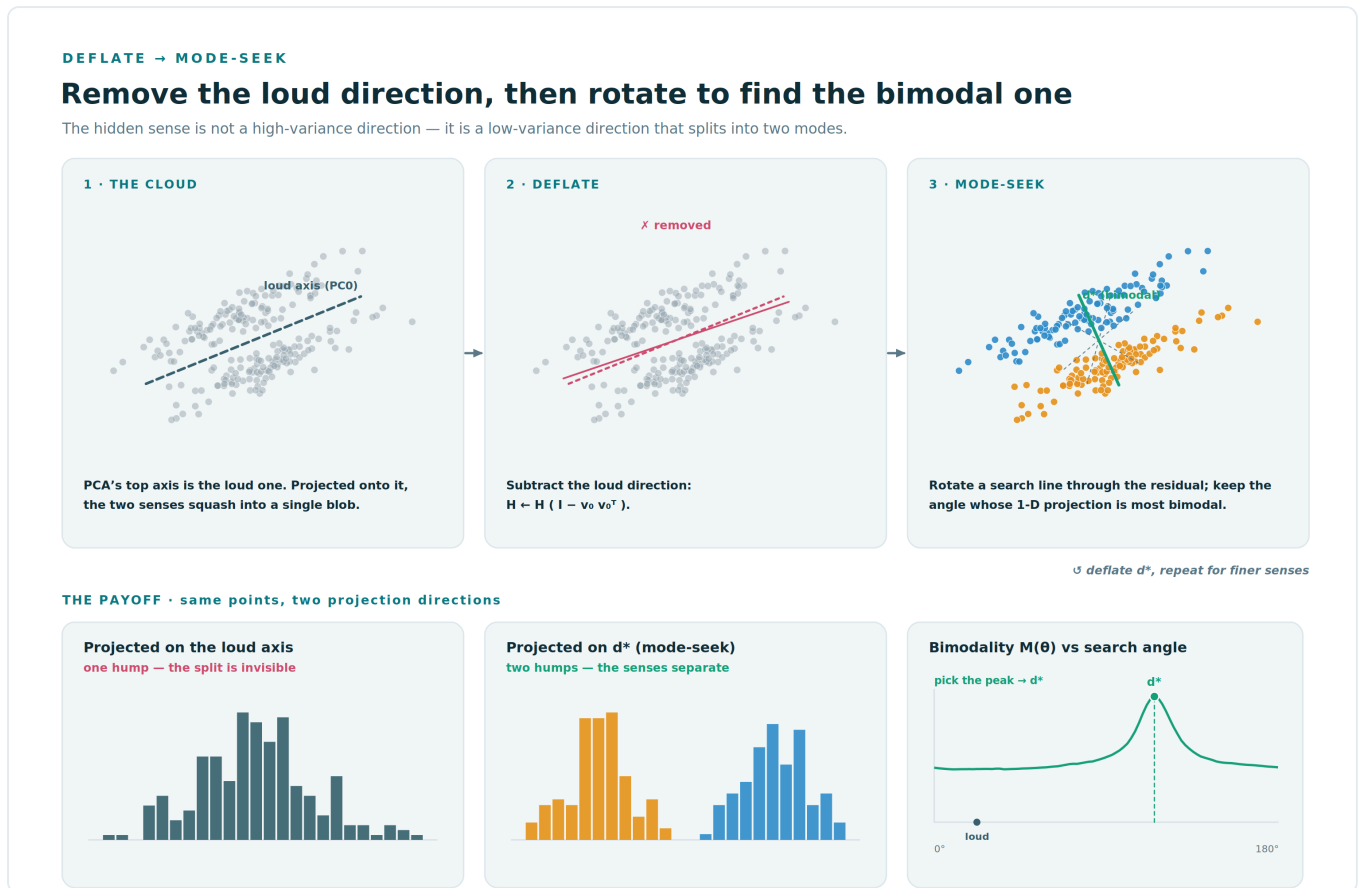


Figure 6. Deflate → mode-seeking. Remove the loud axis, then rotate within the residual until the projection splits into two humps; the $M(\theta)$ sweep marks the optimum d^* .

4.4 Selecting r (how much to deflate) and m (how many senses)

Suppression level r . Choose r to maximize recoverability of residual structure rather than by a fixed variance cutoff:

$$r^* = \operatorname{argmax}_r J(r), \quad J(r) = \max_{\|d\|=1, d \in \operatorname{range}(P_r)} M(d)$$

i.e. deflate just enough that the best residual direction is maximally bimodal, with an eigengap / cluster-stability guard against over-deflation. **Sense count m** is set by the same evidence gap: keep extracting modes while $\Delta\text{BIC} > 0$ (equivalently $\text{BC} > 0.555$); stop when an additional split is no longer supported.

4.5 Recursion — building the hierarchy

After extracting d^* , deflate it as well and repeat, peeling off senses one at a time:

```
P ← P ( I - d* d*T )           (orthogonalize the search against found modes)
repeat 4.3–4.4 until ΔBIC ≤ 0
```

This yields an ordered set of hidden directions $D = [d^*_1, d^*_2, \dots]$ — a steering *subspace* for the Multi-region regime, or a single vector for Masked.

4.6 Constructing the steering vector

Once a target mode is isolated, assign the sign with a handful of orienting exemplars or a gloss — **orientation only, O(1) labels, not a training set**. A ladder of constructions, each improving on the last:

```
d0 = μB - μA           baseline centroid difference (dominated – see 3.1)
d1 = P ( μB - μA )       deflated: loud-subspace leakage removed
d*  = mode-seek direction  unsupervised: finds the split without labels
```

In practice d^* coincides (up to sign) with the bimodal axis, so it equals a normalized d^1 *without ever needing the cluster labels* that d^0/d^1 require — that is the point of mode-seeking: the direction and the split are found jointly.

4.7 Inference-time steering and routing

Apply the recovered direction, gated so it fires only in relevant contexts:

```
pi = softmaxk ( -β · || P hi - ck ||2 )   soft assignment to region centroid ck
h' i = hi + α · gi · d̂                     gi = relevant-region gate, d̂ = d*/||d*
```

For safety use, take $\alpha < 0$ to **suppress** a dangerous minority sense that variance-based methods would never have surfaced.

Peel senses one at a time, then route the steer

Deflate each found direction and repeat; at inference, gate the push to the relevant contexts.

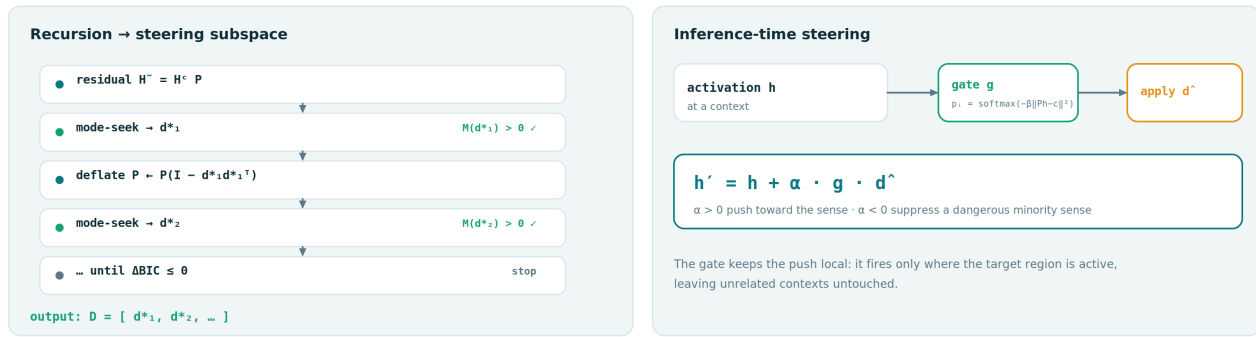


Figure 7. Recursion builds an ordered steering subspace; at inference a context-gate keeps the push local, leaving unrelated contexts untouched.

5 Worked geometry

A controlled construction makes the failure and recovery explicit. Three context groups for a polysemous token, with the institution sense $\sim 3\times$ more frequent than the river sense: the institution sense dominates the variance spectrum (loud axis, top PC), while the river sense lives in a low-variance but bimodal residual direction.

Method	Senses recovered	Why
Difference-of-means	1 / 2	river averaged into institution centroid
Top principal component	1 / 2	variance peak = loud axis; river invisible
Deflate → mode-seek (ours)	2 / 2	suppress loud PC, then cluster the bimodal residual

The bimodality index along the recovered direction lands near $BC \approx 0.7\text{--}0.8$ (clearly two modes) versus $BC \approx 0.33$ along the loud axis (one hump) — a clean, reproducible separation.

RESULT

Recovery: standard methods 1/2, ours 2/2

Same activations; only the variance-hierarchy method surfaces both senses.

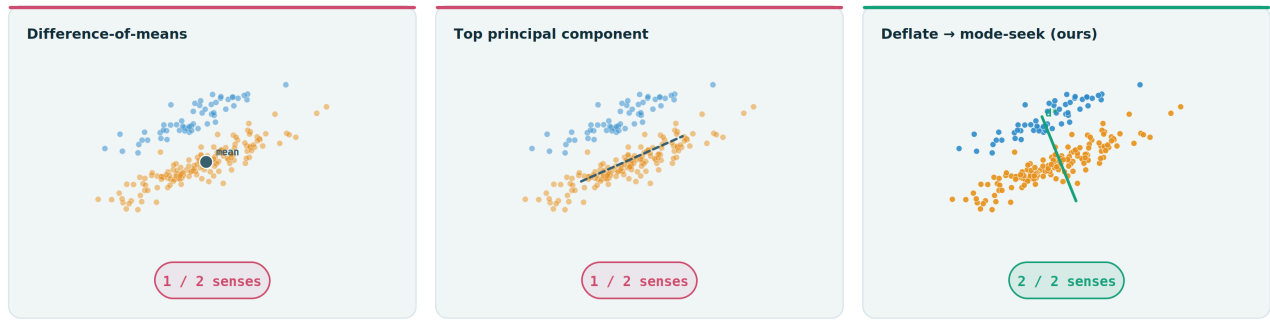


Figure 8. Recovery on identical activations: standard methods surface 1 of 2 senses; the variance-hierarchy method surfaces both.

6 Algorithm

Input: activations $H \in \mathbb{R}^{n \times d}$ for target concept

Output: ordered hidden directions D , with steering gates

```

 $\mu \leftarrow \text{mean}(H)$ ;  $H_c \leftarrow H - \mu$ 
 $\Sigma \leftarrow (1/n) H_c^T H_c$ ;  $\{(\lambda_k, v_k)\} \leftarrow \text{eig}(\Sigma)$ 
 $r^* \leftarrow \text{argmax}_r J(r)$  # 4.4: how much to deflate
 $P \leftarrow I - V_{\{r^*\}} V_{\{r^*\}}^T$  # 4.2: deflation projector
 $D \leftarrow []$ 
repeat # 4.5: recursion
     $d^* \leftarrow \text{argmax}_{\{||d||=1, d \in \text{range}(P)\}} M(d)$  # 4.3: mode-seek (GMM evidence gap)
    if  $M(d^*) \leq 0$ : break # no further mode supported
    orient sign of  $d^*$  with  $O(1)$  exemplars # 4.6
     $D.append(d^*)$ 
     $P \leftarrow P (I - d^* d^{*T})$ 
return  $D$ , routing gates  $\{p_i\}$  # 4.7

```

7 Positioning and defensible novelty

Classical building blocks (not claimed as novel). PCA, projection pursuit, ICA, subspace deflation, and Gaussian-mixture model selection are all textbook. Re-stating $\mu_B - \mu_A$ in heavier notation adds no novelty and, in a patent, is a liability — an examiner strips the notation and asks what the method actually *does* differently.

Where the defensible contribution sits. The novelty argument should rest on the integrated mechanism, not any single equation:

- The **variance-hierarchy diagnosis** and four-regime taxonomy as an explicit router for which recovery procedure to run.

- **Targeted deflation driven by recoverability** (r^* chosen to maximize residual bimodality), not by a fixed variance cutoff.
- **Mode-seek-for-steering-vector construction**: a multimodality objective — not variance, not a supervised discriminant — to build an unsupervised steering direction for a minority sense.
- The **evidence-gap selection rule** unifying direction-finding, existence-of-a-second-mode, and the recursion stopping criterion in one statistic.
- **Recursion + routing** to produce a steering subspace with context-gated application.

The combination — diagnose → deflate-by-recoverability → mode-seek → evidence-gap selection → recurse → route, applied to unsupervised steering — is the candidate inventive step. **Whether it clears novelty / obviousness is a legal determination for patent counsel**; this section is positioning, not a legal opinion.

A Appendix — symbol reference

H, H_c	activations (raw, centered)	P	deflation projector $I -$
Σ	covariance of activations	$\tilde{H} = H_c P$	residual activations
λ_k, v_k	eigenvalue / eigenvector of Σ	$z_i(d)$	1-D projection $d^\top \tilde{h}_i$
V_r	top- r dominant subspace	$M(d)$	multimodality index (BC)
$\Delta = \mu_B - \mu_A$	centroid difference	d^*	recovered hidden direct
$\alpha, \hat{d}$	steering strength, unit vector	g_i, p_i	routing gate / soft ass